

# **DYNAMIC TERRALINK ALGORITHM**

**Algorithm: Dynamic Off-Grid Emergency Communication and Self-Healing Routing**

## **Input**

- Local SOS/user event
- Incoming LoRa packet
- GPS location
- Operator command
- Node configuration
- Available neighbour/next-hop information

## **Output**

- Delivered SOS/message
- Forwarded packet
- ACK / MSGACK / relevant response
- Updated incident/message status
- Recovered communication route when the current path becomes unavailable

### **1. System Initialization:**

1. START
2. Initialize the microcontroller and required peripherals.
3. Initialize the LoRa communication module.
4. Initialize node-specific interfaces:
  - Sender → GPS, OLED, SOS input, buzzer, controls
  - Relay → LoRa interface and routing/packet processing
  - RX1 → LoRa gateway and UART interface
  - RX2 → keypad, UART, local command-center services
5. Initialize packet-processing state:
  - Sequence information
  - Packet/message identifiers
  - Duplicate tracking
  - Node configuration
6. Verify LoRa initialization.
7. IF LoRa initialization fails:  
    Retry initialization.
8. ELSE:  
    Enter continuous monitoring.

## **2. Continuous Monitoring:**

9. Monitor the node continuously.
10. Check for:
  - Local SOS event
  - Local message/input
  - Incoming LoRa packet
  - Required response/acknowledgement
  - Routing/forwarding activity
  - Command-center request
11. IF a local event occurs:  
Go to LOCAL PACKET GENERATION.
12. IF an incoming packet is received:  
Go to PACKET RECEPTION AND VALIDATION.
13. Otherwise:  
Continue monitoring.

## **3. Local SOS / Message Generation:**

14. Detect local event.
15. Determine event type.
16. IF event is SOS:  
Acquire or update GPS location.
17. Generate the required sequence/message identifier.
18. Construct the protocol packet.
19. Populate applicable packet fields:
  - SRC
  - DST
  - SEQ
  - TTL
  - HOP
  - NEXT
  - FROM
  - TYPE
  - MSGID
  - LAT
  - LON
  - MSG
20. Validate the generated packet.
21. Determine an appropriate next hop.
22. Check packet TTL.
23. IF TTL is invalid/expired:  
Drop or terminate packet processing

24. ELSE:  
Update the required hop/lifetime information.
25. Transmit the packet through LoRa.
26. Proceed to acknowledgement/response handling when required.
27. Return to monitoring.

#### **4. Packet Reception and Validation:**

28. Receive LoRa packet.
29. Parse the received packet.
30. Validate packet format and required fields.
31. IF packet is invalid:  
Drop packet.  
Return to monitoring.
32. Check packet identity using the relevant sequence/message identifier.
33. IF packet is already processed:  
Treat it as a duplicate.  
Drop packet.  
Return to monitoring.
34. ELSE:  
Record/process the packet identity.
35. Check TTL.
36. IF TTL has expired:  
Drop packet.  
Return to monitoring.
37. ELSE:  
Update hop/lifetime information.
38. Check packet destination.

#### **5. Destination Decision:**

39. IF DST corresponds to the current node:  
Process the packet locally.  
Determine packet TYPE.  
Execute the appropriate operation:
  - SOS → emergency processing
  - MSG → message processing
  - ACK → acknowledgement processing
  - MSGACK → message-delivery processing
  - LOCREQ → location-request processing
  - LOC → location-response processing

Generate the required response, if applicable.

Forward relevant information toward the appropriate receiver/application layer.

Return to monitoring.

40. ELSE:

The current node is an intermediate node

Determine the next forwarding hop.

Go to NEXT-HOP SELECTION.

## **6. Next-Hop Selection:**

41. Examine available forwarding options.

42. Determine whether a usable next hop exists.

43. IF a valid next hop is available:

Set/select the next hop.

Forward the packet.

Continue response/forwarding handling.

44. IF no usable next hop is available:

Initiate dynamic route recovery.

## **7. Dynamic Self-Healing Route Recovery**

45. Detect next-hop/link unavailability.

46. Mark the unavailable next hop as unusable  
for the current forwarding operation.

47. Search for an alternative available neighbour/  
forwarding node.

48. Alternative next hop available?

IF NO:

Retry according to the communication policy  
or terminate/drop the packet when its forwarding  
conditions cannot be satisfied.

IF YES:

Select the alternative next hop.

49. Update the forwarding information.

50. Re-check packet lifetime/TTL.

51. IF TTL remains valid:

Forward the packet through the alternative route.

52. ELSE:

Drop the packet.

53. Continue normal network operation.

## **8. Acknowledgement and Response Handling:**

54. After a transmission requiring a response,  
monitor for the corresponding response.
55. Determine received response type.
56. IF valid ACK is received:  
Confirm the relevant transmission.
57. IF valid MSGACK is received:  
Mark the message as delivered/acknowledged.
58. IF a location response is received:  
Process the returned location.
59. IF the expected response is not received within  
the applicable communication condition:  
Consider retry or route-recovery processing  
where supported.
60. Return to normal packet/network processing.

## **9. Location Request and Response:**

61. IF a location request is generated:  
Identify the target node.
62. Generate the applicable location-request packet.
63. Forward the request through the LoRa network.
64. Target node receives the request.
65. Obtain available GPS/location information.
66. Generate the location response.
67. Return the location response through the network.
68. RX1 receives the location information.
69. Pass relevant information to RX2.
70. Update the Rescue Command Center.

## **10. RX1 Gateway Processing:**

71. RX1 receives a valid LoRa packet.
72. Validate/process the packet.
73. Determine the packet type and relevant information.
74. Process:
  - SOS information
  - Messages
  - Acknowledgements
  - Location information
  - Relevant network responses
75. Forward processed information to RX2  
through the UART interface.

76. Continue monitoring the LoRa network.

**11. RX2 Rescue Command Center:**

77. RX2 receives information from RX1.

78. Update the local command-center information.

79. Update applicable:

- Incident information
- Active SOS status
- Message status
- Node information
- Network/communication status

80. IF an active SOS is received:

Prioritize the emergency incident.

81. Display/process the emergency information  
for the operator.

82. Check for an operator command.

83. IF no command is generated:

Continue monitoring.

84. IF an operator command/message is generated:

Create the appropriate outgoing packet/command.

85. Assign destination and required protocol information.

86. Send the command through the LoRa network.

87. Monitor the resulting response/acknowledgement.

88. Update the command/message delivery status.

**12. Continuous Network Operation:**

89. Complete the current packet/event operation.

90. Maintain relevant packet-processing state.

91. Continue monitoring incoming packets,  
local events and communication activity.

92. IF another event or packet occurs:

Repeat the corresponding processing branch.

93. ELSE:

Continue monitoring.

94. Repeat indefinitely while the node is operating.

## **ALGORITHM: DYNAMIC TERRALINK COMMUNICATION AND SELF-HEALING ROUTING**

START

1. Initialize MCU, LoRa and node-specific peripherals.
2. Initialize packet, sequence and duplicate-tracking state.
3. Verify LoRa initialization.
4. If initialization fails, retry; otherwise enter monitoring
5. Continuously monitor local events and incoming packets.
6. If a local SOS/message event occurs:
  - a. Acquire GPS information when required.
  - b. Generate sequence/message identifier.
  - c. Construct and validate the packet.
  - d. Select the next hop.
  - e. Check TTL.
  - f. If TTL is valid, update hop/lifetime information and transmit; otherwise drop the packet.
7. If a LoRa packet is received:
  - a. Validate packet format.
  - b. If invalid, discard it.
  - c. Check for duplicate packet identity.
  - d. If duplicate, discard it.
  - e. Check TTL.
  - f. If expired, discard it.
  - g. Update hop information.
  - h. Check destination.
8. If the current node is the destination:
  - a. Process the packet according to TYPE.
  - b. Handle SOS, MSG, ACK, MSGACK, LOCREQ or LOC as applicable.
  - c. Generate the required response.
9. If the current node is not the destination:
  - a. Determine the next forwarding hop.
  - b. If a valid next hop exists, forward the packet.
  - c. If the next hop is unavailable, initiate route recovery.
10. During route recovery:
  - a. Avoid the unavailable next hop.
  - b. Search for an alternative forwarding node.
  - c. If an alternative is available, update the next hop and forward the packet.

- d. Otherwise retry or terminate forwarding according to the applicable condition.
- 11. Process required ACK/MSGACK/location responses.
- 12. RX1 processes received network information and passes relevant data to RX2.
- 13. RX2 updates the Rescue Command Center and handles operator-generated commands/messages.
- 14. Return to continuous monitoring.

END